

Numerical calculation with experimental validation of pressure drop in a fixed-bed reactor filled with the porous elements

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Abstract

To study flow dynamics in a fixed-bed reactor, experiment and numerical simulation are used. In this work, the flow dynamics in a fixed-bed reactor filled with porous particles was simulated. For verification, experimental data were used. Porous particles were prepared from grains with sizes from 0.2 to 2.0 mm. Porous particles made by sintering grains in a muffle furnace. The study of pressure drop was performed in the velocity range up to 3 m/s. Numerical calculation was performed for a realistic computational domain obtained by RBD-algorithm (rigid body dynamics). It was found that if the pore size is less than 0.5 mm, then the flow through the porous medium of particles is minimal; if the pore size is larger than 0.5 mm, then there is a flow through the porous elements of the fixed-bed reactor. To take into account the porosity of the particles, γ correlation coefficient was proposed for the Ergun equation.

KEYWORDS

CFD modeling, experiment, fixed-bed, porous medium

1 | INTRODUCTION

The devices with a packed bed filled the particles are widely used in various industries: chemical engineering,^{1,2} energy,³ petrochemistry,⁴ and so on. The packed beds are used to improve the contact between two phases for heat and mass transfer in chemical reactors, in absorbers, and in thermochemical recuperators,⁵⁻⁷ and so on. The calculation of the gas dynamics of the flow in such devices is an important engineering and scientific-technical problem. Despite the relatively simple arrangement of the packed beds, in which gas flows through a tube filled with the particles, it has proven to be no simple matter to obtain reliable estimates of flow dynamic parameters such as the pressure drop, outlet velocity, and so on. One possible reason is the complex flow path for the fluid, due to the random arrangement of the particles of the packed beds, which are themselves of many different sizes and shapes, according to the needs of the particular process.

There are several approaches to the study of the flow dynamics in a packed bed filled with various particles. The three main approaches are experimental research,^{8,9} numerical simulation,¹⁰⁻¹² and analytical calculation based on the use of semiempirical expressions^{13,14} to determine the pressure drop in a packed bed. The first

method is based on the instrumental determination of flow characteristics at the packed bed outlet. For example, the outlet velocity and the pressure drop in the packed bed. The second method is based on developing a packed bed CFD model and conducting a numerical study, which also determines the velocity, pressure drop, and other flow characteristics. Finally, the third method is based on the use of various empirical and semiempirical expressions to determine the pressure drop in a packed bed. The variety of particle shapes, operating modes of the packed bed devices, physicochemical properties of substances leads to the fact that the scientific literature has a wide variety of modified equations for determining the pressure drop in the packed beds.¹⁵⁻¹⁷ The traditional approach to the analytical calculation of pressure drop in the fixed-bed apparatus filled with particles of various shapes is based on the Ergun equation¹³ and its various modifications¹⁸⁻²¹:

$$\frac{|\Delta p|}{L} = \frac{150\mu(1-\epsilon)^2}{D_p^2 \epsilon^3} v_s + \frac{1.75\rho(1-\epsilon)}{D_p^2 \epsilon^3} v_s^2 \quad (1)$$

where $|\Delta p|$ is the pressure drop in the packed bed for the length L ; μ is the viscosity; ρ is the density; v_s is the superficial velocity (i.e., the

velocity that the fluid would have through the empty tube at the same volumetric flow rate); ϵ is the void fraction of the bed (interparticle porosity); D_p is the mean particle diameter.

With the increasing computing power of computers, numerical methods are becoming increasingly popular for studying the flow dynamics in the packed beds.^{22–24} For CFD simulations, various special software products are used, such as, for example, ANSYS,^{25–27} Comsol,^{28,29} OpenFoam,^{30,31} and others. A distinctive feature of the numerical simulation of a packed bed is the calculation complexity because the computational domain consists of many particles. At the same time, there are a number of scientific articles in the literature dealing with numerical modeling of flow dynamics in the packed beds.

Bai et al¹¹ published the results of experimental and numerical investigation of the pressure drop in a packed bed for different values of tube-to-particle diameter ratios. For numerical modeling, the discrete element method (DEM) and computational fluid dynamics (CFD) are combined to simulate a packed bed with low tube-to-particle diameter ratios ($D/d_p < 4$). The DEM is used to develop a randomly packed bed with particles of various shapes. The predicted pressure drops are in good agreement with the experimental results with errors less than the desired limit (10%) for the industrial design of the packed bed reactors. However, despite the fact that the authors use porous particles, in a numerical model, the particles are defined as solid bodies. The porosity of the particles was not taken into account.

Calis et al¹⁰ also studied the packed bed filled with catalysts of various shapes. The main attention is paid to numerical modeling of flow dynamics in a packed bed. In that article, a commercial CFD code (CFX-5.3) was used to determine, with an average error of about 10%, the pressure drop characteristics of packed beds filled with the spheres that have a tube-to-particle diameter ratio of 1.00–2.00. In this work, the catalyst particles are also represented as the solid bodies, and the real porosity of the catalyst is not taken into account.

Lao et al considered a steam methane reforming reactor as a porous medium.³² The authors presented the results of CFD model of an industrial-scale reforming tube is developed in ANSYS Fluent with realistic geometry characteristics to simulate the transport and chemical reaction phenomena with approximate representation of the catalyst packing. To estimate the pressure drop across the porous zone in the CFD modeling of turbulent flows, a semiempirical expression, the Ergun equation, which is applicable over a wide range of Reynolds numbers and for many packing patterns is used.

Some authors claim that the porosity of the particles does not have a significant effect on the pressure drop in a packed bed. There is a work presented by Dixon and co-authors, in which CFD modeling considers a packed bed filled with porous catalyst particles. In the CFD model, the authors set the porosity of the particle. The authors showed that the flow through the porous medium of catalyst particles is much less than the flow through the flow area between the catalyst particles.³³ In this work, the type and structure of the catalyst are not indicated, but, most likely, the simulation results are given for the nickel-based catalyst for methane steam reforming. In this article, the authors showed a good correlation between the results of CFD modeling and experimental data.

It is noteworthy that 10–20 years ago, the authors worked with the packed beds consisting of a small number of particles for numerical simulation. This is due to the fact that CFD modeling of the flow in a fixed bed requires large computational powers. Nowadays, modern computing power allows us to build computational domains consisting of several thousand particles. In this case, using the RBD (rigid body dynamics) approach,³⁴ it becomes possible to create realistic packed bed structures.

Many published papers on the study of flow dynamics in a packed bed consider porous catalysts as particles, in which the pore size does not exceed 0.1 mm. At the same time, when the authors of this article conducted research to determine the pressure drop in a fixed bed filled with particles with a pore size of 0.5–2 mm, the experimental results always differed from the results obtained using the Ergun equation, and the results of CFD modeling, in which particles are treated as solids. In this article, an experimental study and numerical modeling of flow dynamics in a packed bed filled with porous particles was performed. The conditions are determined under which the porous particles of a packed bed can be considered as solids and the conditions under which the porosity of the particles must be taken into account. More effort has been made in comparing the results of numerical modeling, experimental data, and results obtained using the Ergun equation. Moreover, experimental and numerical research was performed for a large number of elements (more than 1,000) and a realistic RBD-algorithm was used to generate the computational domain, which ensures high accuracy of the results.

2 | EXPERIMENT

To obtain calibration values of the pressure drop in a packed bed filled with porous particles, a series of experiments were performed. The photograph of the experimental setup is shown in Figure 1. The experimental setup consists of the following main elements: (a) an

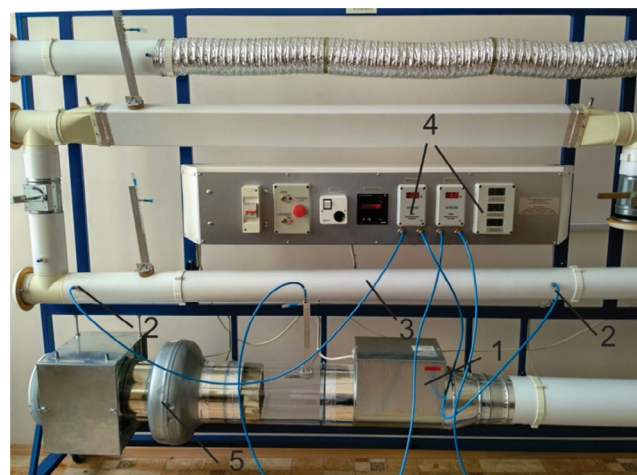


FIGURE 1 A photograph of an experimental setup for investigation of fluid dynamics and pressure drop of a packed bed [Color figure can be viewed at wileyonlinelibrary.com]

electric heater; (b) the flexible manometer tubes; (c) a cylindrical tube filled with porous particles; (d) a dashboard; (e) a fan.

The working medium is atmospheric air. The flow rate and velocity of air are controlled by changing the fan speed on the dashboard (d). The pressure is measured using differential pressure gauges with digital indication of readings. To obtain stable values of the pressure, the pressure measurement points are located at a distance of three diameters before the inlet and one diameter after the outlet. All experiments were conducted at ambient temperature ($\approx 20^\circ\text{C}$).

The packed bed is filled with cylinders (20×20 mm) and cubes (20×20 mm). The particles were prepared by using crashed $\alpha\text{Al}_2\text{O}_3$ -support of the Ni-based catalysts. The particles of a crashed $\alpha\text{Al}_2\text{O}_3$ -support have a size of 0.2–2 mm. After crushing, small elements are separated using a sieve: 0.5, 1.0, 1.5, 2 mm. It also used quartz sand with a particle size of about 0.2 mm. A muffle furnace is used to produce the particles of the packed bed. The grains of a crashed $\alpha\text{Al}_2\text{O}_3$ -support were sintered in a muffle furnace at a temperature of $1,200^\circ\text{C}$ for 3 min. A schematic representation of the producing of the cylinders and cubes of the packed bed is shown in Figure 2.

Characteristics of the particles after sintering in the furnace are shown in Table 1. Table 1 presents the porosity of the particles and the porosity of the packed beds filled with these particles. After sintering the particles were placed in a cylindrical tube with a diameter of 80 mm. The length (depth) of the packed bed is 800 mm. The packed beds were supported by four pins that passed through the cross-section of the column. The author chose the length-to-diameter

(L/D) about 10 to get more accurate values of the pressure drops. The packed bed structure is chosen in such a way that the interparticle porosity for each packed bed was equal for each grain size. Interparticle porosity for cylinders is $\epsilon_p = 0.33$, for cubes $\epsilon_p = 0.39$.

The in-particle porosity was measured via a helium porosimetry method. This method relies on the expansion of helium gas. The schematic diagram of the helium porosimetry method is presented in Figure 3. The pressure drop in a helium porosimetry system depends on how much space there is in Chamber 2, and that depends on how much of Chamber 2 is occupied by solid grains.

3 | CFD MODELING

3.1 | Geometry

The computational domain (geometry) is generated in such a way that the real structure of the packed bed is repeated as accurately as possible. The main criteria for the similarity of the computational domain and the real packed bed are the number of catalyst particles and the depth (length) of the packed bed. For all computational domains, the number of particles and the length of the packed bed is equal to the values for a real object. For all shapes of particles (cylinders and cubes), the length and diameter of the packed bed are constant. The computational domain of the packed bed for CFD modeling is shown in Figure 4.



FIGURE 2 A schematic representation of producing of the cylinders and cubes for the packed bed: (a) the grains of the crashed $\alpha\text{-Al}_2\text{O}_3$ -support of the Ni-based catalysts; (b) a muffle furnace; (c) the cylinders and cubes for the packed bed [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Characteristics of the particles and the packed beds after sintering in the furnace

		Grain size, mm				
		0.2	0.5	1.0	1.5	2.0
Porosity of particle/(in-particle porosity)		0.39	0.40	0.42	0.46	0.52
Porosity of packed bed/(interparticle porosity)	Cylinder	0.33	0.33	0.33	0.33	0.33
	Cube	0.39	0.39	0.39	0.39	0.39

The RBD-approach (rigid body dynamics) is used to generate the computational domain, which allows us to create a randomly filled structure. The RBD-approach is often used in the field of graphic design and modeling. The Blender software is used as a platform for RBD modeling. The packed bed generated in Blender is filled randomly based on Newton's laws of motion and Lagrange mechanics.

3.2 | Mesh

The mesh for the computational domain (Figure 4) was generated via ANSYS Meshing. The total number of elements of the mesh for all cases studied is about 20 million. The number of elements of the mesh is determined after the grid independence study, where pressure and flow rate at the packed bed outlet are chosen as the control parameter. The mesh structure of the packed bed and interparticle space (flow area) is shown in Figure 5. The elementary cells of the mesh are tetrahedra. For the flow area near the particle walls, the mesh was thickened to correctly simulate the boundary layer. For this

purpose, the inflation function with the number of layers equal to five was used.

Meshing the packed bed with randomly filled particles is inherently complex due to the presence of a high number of particle-particle and particle-wall contact points. To overcome this difficulty, the inflations were added for all particles and walls. Moreover, the contact points were set as the contact regions.

Due to the complexity of the mesh as a result of the developed surface, the scalable wall function (SWF) was used to calculate the boundary layer. The mesh was adapted to the dimensionless distance from the first grid node to the wall (y^+). Adaptation is made in such a way that the condition is satisfied— $y^+ < 50$. For the main part of the flow area, the parameter y^+ does not exceed 25, only in the areas of contact of the particle surfaces with each other and with the pipe wall $25 < y^+ < 50$. The main purpose of using SWF near walls is to force the use of a logarithmic law in combination with the standard calculation. This is achieved by introducing a constraint in the calculations of y^+ in such a way that the following condition is satisfied:

$$\bar{y}^+ = \text{MAX}(y^+; y_{\text{lim}}^+) \quad (2)$$

where y_{lim}^+ is maximum allowed value for each element of the mesh.

To adapt the mesh for y^+ , the procedure *adapt* in Fluent solver was used. The *inflation* function is used because when using a tetrahedral mesh, the procedure for adapting the near-wall nodes leads to irregular shredding of the nodes and, therefore, to an uneven boundary layer. The inflation function creates hexahedral cells at the wall, which, when crushed, also remain hexahedra, which allows obtaining the correct boundary layer.

In the computational domain shown in Figure 4, the particles are represented as a porous medium. In this regard, the porous particles are the flow area of the computational domain. To avoid the intersection of the flow areas represented by the interparticle flow area

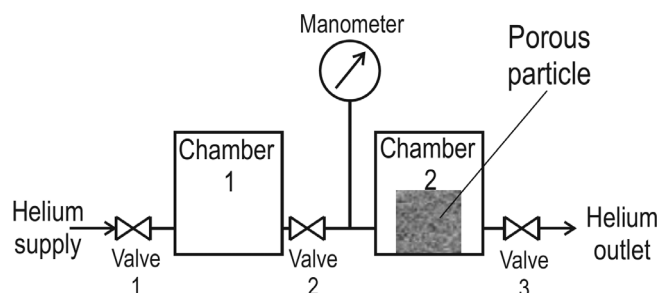


FIGURE 3 A schematic representation of helium porosimetry method

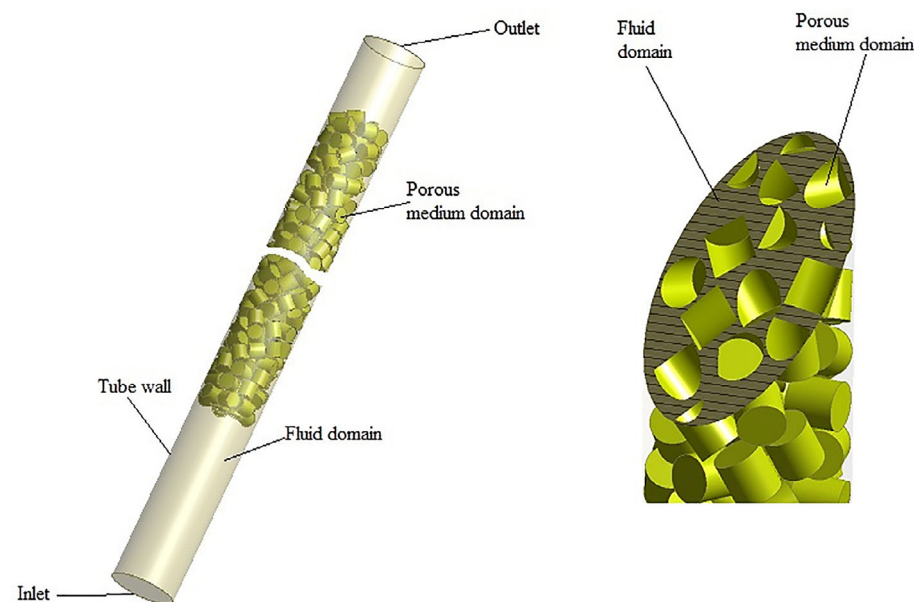


FIGURE 4 Computational domain of the packed bed filled with the cylinders 20 × 20 mm [Color figure can be viewed at wileyonlinelibrary.com]

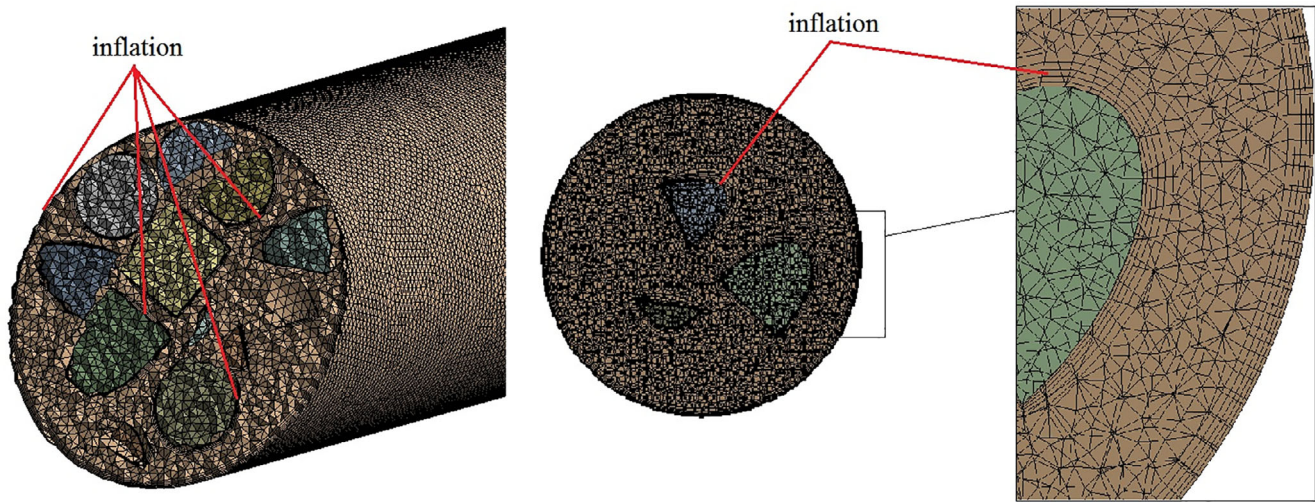


FIGURE 5 The mesh structure for the computational domain presented in Figure 4 [Color figure can be viewed at wileyonlinelibrary.com]

and flow areas represented by the porous medium flow areas, the contact region was set through the setup of the corresponding interfaces. For this purpose, the method of combining into an integral part (*part*) is used, which only works if there are *coupled-wall* interfaces in the computational domain.

The mesh structure is shown in Figure 5.

3.3 | Governing equations

In presented CFD model the conservation and momentum equations in combination with source term for a porous medium equation are solved in ANSYS Fluent. The assumptions are taken for CFD modeling: steady-state conditions; the temperature of flow is constant; no the flux of energy due to a mass concentration gradient (no Dufour effects³⁵); work by viscous forces and by pressure is not done; surface oxidation reactions of the metal wall are absent.

The mass conservation equation (the continuity equation) for steady-state flow can be written as follows:

$$\nabla \cdot (\rho \vec{v}) = 0 \quad (3)$$

For 3D computational domain, the continuity equation can be written as:

$$\frac{\partial}{\partial x}(\rho v_x) + \frac{\partial}{\partial y}(\rho v_y) + \frac{\partial}{\partial z}(\rho v_z) = 0 \quad (4)$$

In these equations v_x , v_y , and v_z are the velocity components; x , y , and z are the axial and radial coordinate, respectively.

Conservation of momentum in an inertial (nonaccelerating) reference frame is described by follow equations³⁶:

$$\nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla \cdot p + \nabla \cdot (\bar{\tau}) + \rho \vec{g} + \vec{F} \quad (5)$$

where p is the static pressure; $\rho \vec{g}$ and $\vec{F}$ are the gravitational body force and external body forces, respectively. The stress tensor $\bar{\tau}$ is given by follows equation:

$$\bar{\tau} = \mu \left[(\nabla \vec{v} + \vec{v}^T) - \frac{2}{3} \nabla \cdot \vec{v} I \right] \quad (6)$$

where μ is the molecular viscosity, I is the unit tensor, and the second term on the right hand side is the effect of volume dilation.

The processes in the packed beds are taking place at the wide ranges of velocity. The flow around the particles may have a vortex after the particle. Therefore in the CFD model the RNG $k-\epsilon$ model is chosen.

$$\frac{\partial}{\partial x_i}(\rho k v_i) = \frac{\partial}{\partial x_j} \left(\alpha_k \mu_{\text{eff}} \frac{\partial k}{\partial x_j} \right) + G_k + G_b - \rho \epsilon - Y_M + S_k \quad (7)$$

and

$$\frac{\partial}{\partial x_i}(\rho \epsilon v_i) = \frac{\partial}{\partial x_j} \left(\alpha_\epsilon \mu_{\text{eff}} \frac{\partial \epsilon}{\partial x_j} \right) + C_{1\epsilon} \frac{\epsilon}{k} (G_k + C_{3\epsilon} G_b) - C_{2\epsilon} \rho \frac{\epsilon^2}{k} - R_\epsilon + S_\epsilon \quad (8)$$

In Equations (7) and (8) there are following terms:

- G_k —the generation of turbulence kinetic energy due to the mean velocity gradients.
- Y_M —the contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate.
- G_b —the generation of turbulence kinetic energy due to buoyancy.
- α_k and α_ϵ —the inverse effective Prandtl numbers for k and ϵ , respectively.

The additional term R_ϵ in Equation (8) equation can be defined as:

$$R_\epsilon = \frac{C_\mu \rho \eta^3 (1 - \eta/\eta_0) \epsilon^2}{1 + \beta \eta^3} k \quad (9)$$

where $\eta \equiv Sk/\epsilon$, $\eta_0 = 4.38$, $\beta = 0.012$.

The model constants $C_{1\epsilon}$ and $C_{2\epsilon}$ in for the Equations (7) and (8) used by default setup for Fluent solver of ANSYS³⁶:

$$C_{1\epsilon} = 1.42; \quad C_{2\epsilon} = 1.68 \quad (10)$$

Porous media are simulated by the addition of a momentum source term to the momentum Equation (5). The source term is composed of two parts: a viscous loss term, and an inertial loss term. The momentum source term for simple homogeneous porous media can be written as:

$$S_i = - \left(\frac{\mu}{\alpha} v_i + C_2 \frac{1}{2} \rho |v| v_i \right) \quad (11)$$

where α is the permeability and C_2 is the inertial resistance factor; μ is the viscosity; ρ is the density.

The permeability and inertial loss coefficient in each component direction may be determined based on the semiempirical Ergun equation:

$$\alpha = \frac{D_p^2}{150} \frac{\epsilon_{ip}^3}{(1 - \epsilon_{ip})^2} \quad (12)$$

and

$$C_2 = \frac{3.5 (1 - \epsilon_{ip})}{D_p \epsilon_{ip}^3}. \quad (13)$$

In Equations (12) and (13), D_p is the mean particle diameter (diameter of grains), L is the packed bed length, and ϵ_{ip} is the in-particle porosity.

In the solution setup the solution parameters were retained as the default settings of ANSYS Fluent except²⁷:

- Second order upwind for the momentum.
- Least squares cell based for the gradient.

4 | RESULTS AND DISCUSSION

One of the main advantages of using special programs for CFD modeling is the ability to obtain visual contours of various operational parameters. In Figure 6 the velocity contour for the investigated computational domain Figure 4 is shown.

The contour in Figure 6 was obtained at velocity inlet of 0.5 m/s for the particles produced from 0.2 mm grains.

4.1 | Mesh independence study

Five computational meshes were used to determine mesh independence of the CFD model. The results of the mesh independence study presented that good meshes (12 million and 25 million elements) provided almost similar numerical results. Notable divergences were only for the moderate–high velocities.

Clearly, the mesh refinement led to a noticeable change in the pressure distribution in the computational domain. However, the qualitative characteristics of the profile were unaffected except its part for high velocities. Unfortunately, the nature of fluctuations for the high velocities remains unclear.

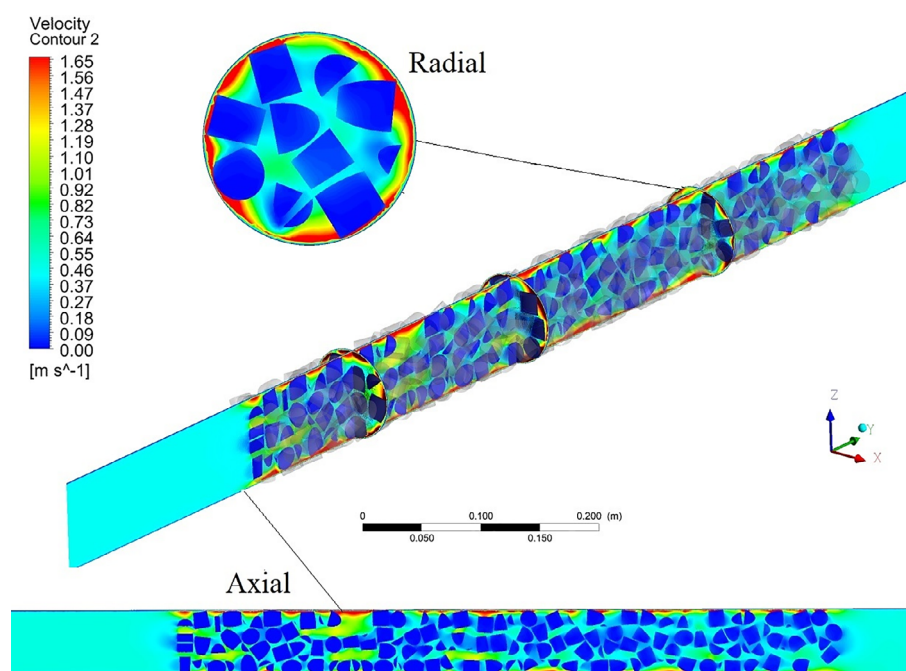


FIGURE 6 The velocity contour for the computational domain presented in Figure 4 [Color figure can be viewed at wileyonlinelibrary.com]

Therefore, it was concluded that the numerical results were independent of mesh resolution when the computational mesh consists of about 12 million elements.

4.2 | Model verification

For the model verification, the pressure drops obtained via various experiments were compared. The following results were compared: the experimental results for the packed bed filled with the particles produced from 0.2 mm grains; the numerical results obtained via CFD modeling with taking into account the porosity of particle; the

numerical results obtained via CFD modeling where the particles were considered as the solid bodies. Dixon et al concluded that flow through the porous particles is insignificant for the particles with small pore size.³⁷ These results were confirmed in our study. Figure 7 shows the comparison of the numerical and experimental results.

Figure 7 depicts a good correlation between the numerical results and experimental data for a wide range of inlet velocity with the divergence of results less than 7–8%. Moreover, the simulation results obtained for the porous and solid particles are almost similar. Therefore, it can be concluded that flow through the particle prepared with 0.2 mm grains is negligible. Other numerical and experimental results presented below also are in a good correlation.

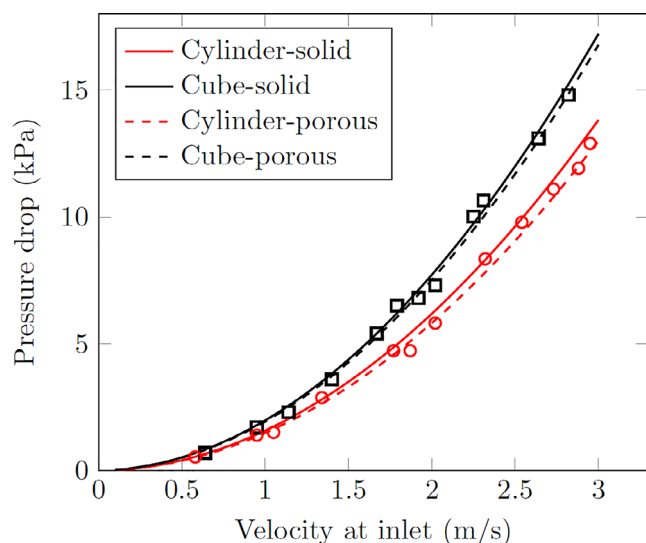


FIGURE 7 Comparison of the numerical results and experimental data: hollow points—experimental results; solid lines—numerical results obtained via CFD modeling with taking into account the porosity of particle; dashed lines—numerical results obtained via CFD modeling where the particles were considered as the solid bodies. CFD, computational fluid dynamics [Color figure can be viewed at wileyonlinelibrary.com]

4.3 | Pressure drop

The experimental results for the pressure drop in the packed bed are shown in Figure 8. For clarity, the results in the range up to 3 m/s the pressure drop are transferred to the logarithmic coordinate plane. Thus, Figure 8a shows a linear coordinate plane, and Figure 8b shows a logarithmic plane. Characteristic quadratic dependences of the form $\Delta P = f(u^2)$ are observed for all the investigated variants.

Figure 8 shows that with an increase in the grain sizes of from which the particles of the packed bed are prepared, the pressure drop decreases. Moreover, for grain sizes for 0.2 and 0.5 mm, the pressure drop is approximately the same. This confirms the fact that for such grains the flow through the particles of the packed bed is minimal. However, as the grain size increases, pressure drop decreases. Moreover, it is noticeable that the decrease in pressure drop with increasing grain size is linear. Therefore, it can be assumed that in porous elements in which pore sizes of more than 0.5 mm, porosity will have a noticeable effect on pressure loss. Therefore, with an increase in pore size 10 times from 0.2 to 2 mm, pressure drop decreases by more than three times. Such an observation allows us to state that the classical Ergun equation for a packed bed can be used only for porous particles in which the pore size is not more than 0.5 mm. With

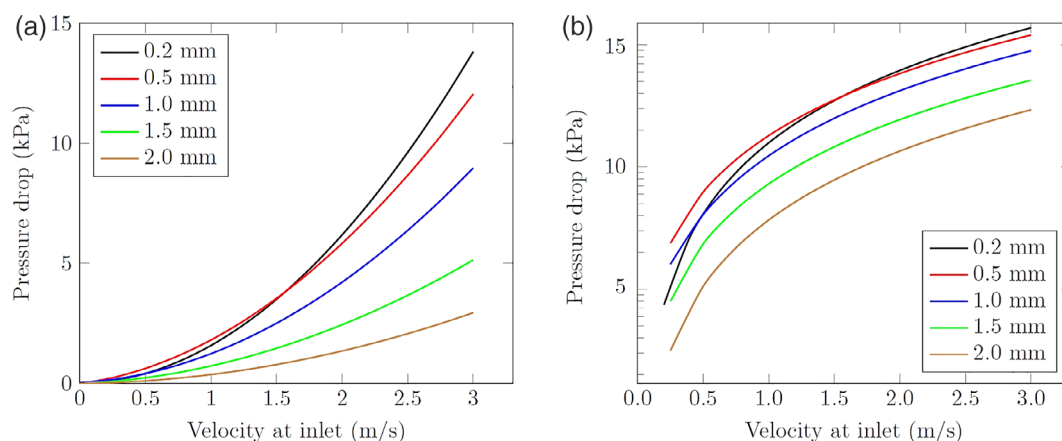


FIGURE 8 The pressure drop in the packed bed on the linear (a) and logarithmic (b) coordinate plane [Color figure can be viewed at wileyonlinelibrary.com]

increasing pore size, it is necessary to add correlation coefficients to the Ergun equation.

One of the most important results of this work is the comparison with the Ergun equation and a key contribution is a simple way to modify one of the parameters of the Ergun equation to account for particle porosity on the pressure drop. When the particles are solid (particle porosity is zero) this parameter has a nominal value and as the particle porosity increases from 0 to 1, this parameter changes as a function of particle porosity and flow velocity. Such an explicit functional dependence of the correlation coefficient in terms of particle porosity and velocity could be obtained from CFD and experiments.

The most successful semiempirical expression describing the laminar flow of Newtonian fluids through packed beds is the Blake–Kozeny equation. This equation is a modification of Ergun Equation (1). The Blake–Kozeny equation is first term of right side of Ergun Equation (1).

$$\frac{|\Delta p|}{L} = \frac{150\mu(1-\epsilon)^2}{D_p^2 \epsilon^3} v_s \quad (14)$$

In this equation the “permeability” of the packed bed can be expressed as follows:

$$k = \frac{150(1-\epsilon)^2}{D_p^2 \epsilon^3}. \quad (15)$$

This expression is valid only if there is no flow through the porous medium of particles of the packed bed. However, as shown in Figure 8, as the pore size of the particles of the packed bed increases, the pressure drop decreases. Therefore, to take into account porosity, it was proposed to add a correlation coefficient γ for expression (12) as follows:

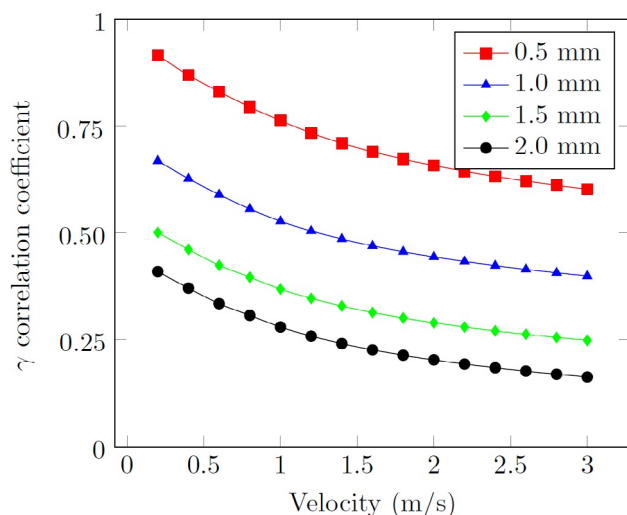


FIGURE 9 The correlation coefficient γ for various pore sizes as a function of velocity [Color figure can be viewed at wileyonlinelibrary.com]

$$k = \gamma \frac{150(1-\epsilon)^2}{D_p^2 \epsilon^3}. \quad (16)$$

To determine the correlation coefficient γ , the results of the experiment and numerical simulation were processed. Using numerical modeling and experiment, it was found that for various forms of particles of the packed beds, the correlation coefficient has close values. However, it was found that the correlation coefficient γ depends not only on pore size (grain size) but also on velocity. In Figure 9 shows averaged values of the correlation coefficient as a function of flow rate for various pore sizes. The correlation coefficient γ has physical meaning. This coefficient shows what gas flow passes through the porous medium of the particles of the packed bed.

5 | CONCLUSION

It has been shown that a commercial CFD code can be used to determine the pressure drop characteristics for the fixed-bed reactors filled with porous particles, with an average error of less than 10%. This error is acceptable for design purposes. To prepare a computational domain of a fixed-bed reactor, a realistic RBD-algorithm (rigid body dynamics) can be used. It was found that if the pore size is less than 0.5 mm, then the flow through the porous medium of particles is minimal and the particles in numerical analysis can be considered as the solid bodies; if the pore size is larger than 0.5 mm, then there is a flow through the porous elements of the fixed-bed reactor. With an increase in pore size 10 times from 0.2 mm to 2 mm, pressure drop decreases by more than 3 times. To take into account the porosity of the particles, a γ correlation coefficient was proposed for the Ergun equation. It was found that γ correlation coefficient depends on pore size and gas velocity. A graphical dependence of γ correlation coefficient on velocity for various pore sizes is obtained. With increasing pore size and gas velocity, γ correlation coefficient decreases.

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